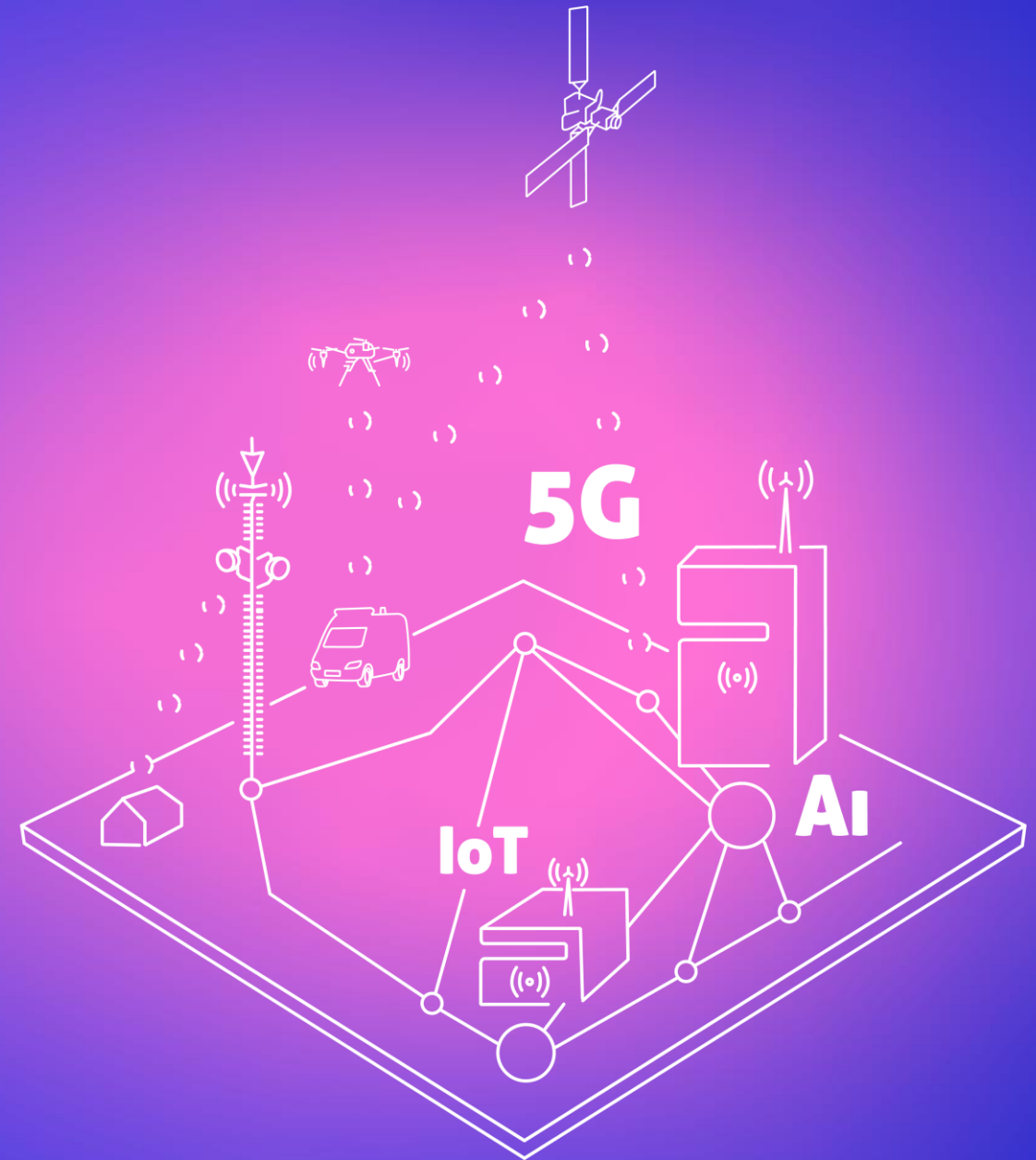




7. RL Introduction

www.cellular-expert.com



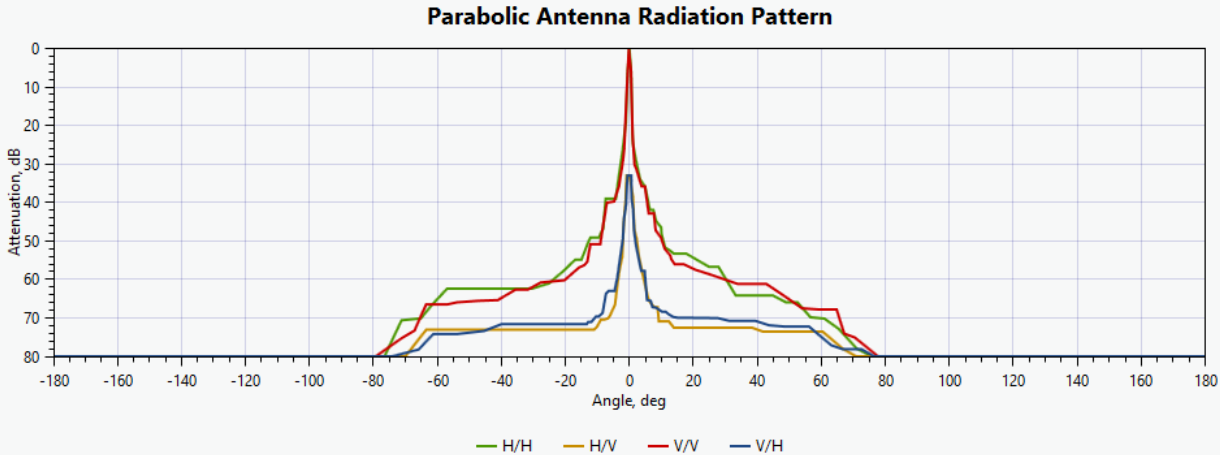
- Antennas > Parabolic
- Radio Models
- Frequency Plans
- Spectrum Mask

Antenna Viewer

Antenna Type: Parabolic

Antenna: UHX8-127

Object ID	Manufactur	Model	Frequency,	Gain, dBi
16	ANDREW CORPOF	UHX8-127	12700	46



Angle, °	H/H	Angle, °	H/V	Angle, °	V/H	Angle, °	V/V
-180	80	-180	80	-180	80	-180	80
-171	80	-171	80	-171	80	-171	80
-169	80	-169	80	-169	80	-169	80
-76.52	80	-70	80	-74.09	80	-79.23	80
-71.21	70.61	-63.5	73.06	-65.89	78.22	-70.94	75.14
-65.17	70.12	-15	73.06	-61.29	74.21	-67.15	73.28
-61.56	66.48	-11.09	73.06	-54.08	74.21	-63.55	66.5
-56.97	62.41	-10.1	72.29	-45.43	73.38	-56.97	66.5
-30.56	62.41	-8.92	70.43	-40.02	71.62	-53.9	65.95
-25.15	61.06	-7.84	70.43	-15	71.62	-47.32	65.57
-20.73	57.88	-6.67	70.12	-13.34	71.62	-41.1	65.37
-16.95	54.89	-5.95	69.23	-12.89	71.05	-35.6	62.64
-15	54.89	-4.51	66.56	-11.9	71.05	-31.73	62.64
-14.87	54.89	-3.24	58.28	-10.46	69.64	-27.76	60.8

Radio Models

Radios

Manager Import

+

▼ Radio: FlexiHopper13_4s_2E1

ID	Model	Power, dBm
1	Default	18
2	FlexiHopper13_4s_2	18

Radio

Modulations

▼ General

Model:
FlexiHopper13_4s_2E1

Manufacturer:
NOKIA

Capacity:
2E1

Emission Designator
3M50G7W

Frequency From, MHz:
12750

Frequency To, MHz:
13250

Bandwidth, MHz:
3.5

MTBF, h:
300000

MTTR, h:
24

Bit Rate, kbps:
2048

Block Size, bits:
2048

Burst Errors:
1

Dispersive Fade Margin, dB:
50

Radios

Manager Import

+

▼ Radio: FlexiHopper13_4s_2E1

ID	Model	Power, dBm
1	Default	18
2	FlexiHopper13_4s_2	18

Radio

Modulations

▼ Receiver

BER 10⁻³ Threshold, dBm
-91

BER 10⁻⁶ Threshold, dBm
-88

Noise Floor, dB:
-108.5365487134953

Noise Figure, dB:
5

Receiver Noise, dBm:
-103.5365487134953

Residual BER:
1E-12

IIP2, dBm:
0

IIP3, dBm:
0

XPIF, dB:
0

Manufacturer's guaranteed minimum XPD, dB:
0

Carrier-to-interference ratio for a reference BER, dB:
0

▼ Transmitter

Power, dBm:
18

Power Low, dBm:
18

Power High, dBm:
0

Radios

Manager Import

+

▼ Radio: FlexiHopper13_4s_2E1

ID	Model	Power, dBm
1	Default	18
2	FlexiHopper13_4s_2	18

Radio

Modulations

▼ Adaptive Modulation:

ID	Technology
1	Microwave

Modulations
QAM0256

▼ QAM0016

Technology:
Microwave

Name:
QAM0016

Sensitivity, dBm:
-73

SNR, dB
15

Code Rate
0

Bps, Hz
0

Downlink Throughput, Mbps
0

Uplink Throughput, Mbps
0

Downlink Bitrate, Mbps
0

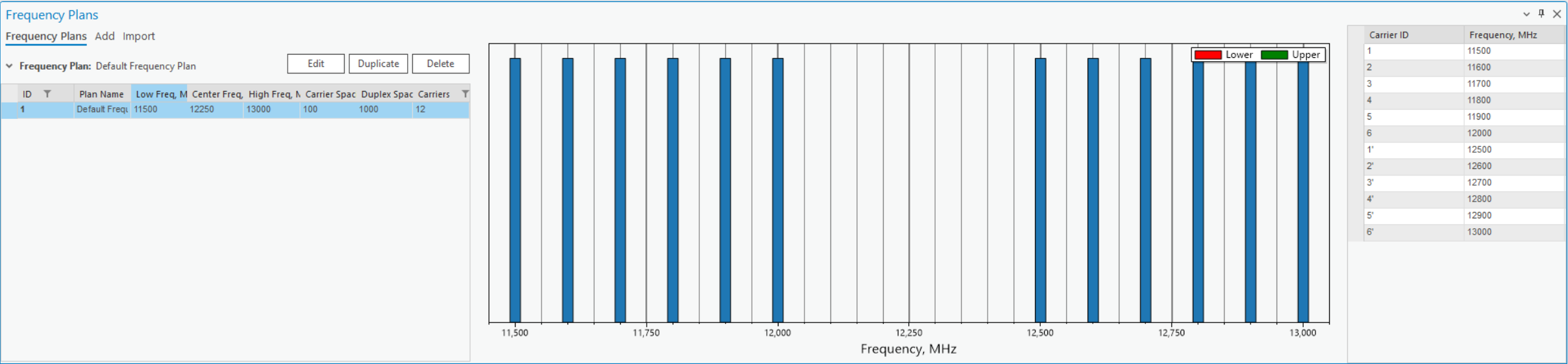
Uplink Bitrate, Mbps
0

> QAM0128

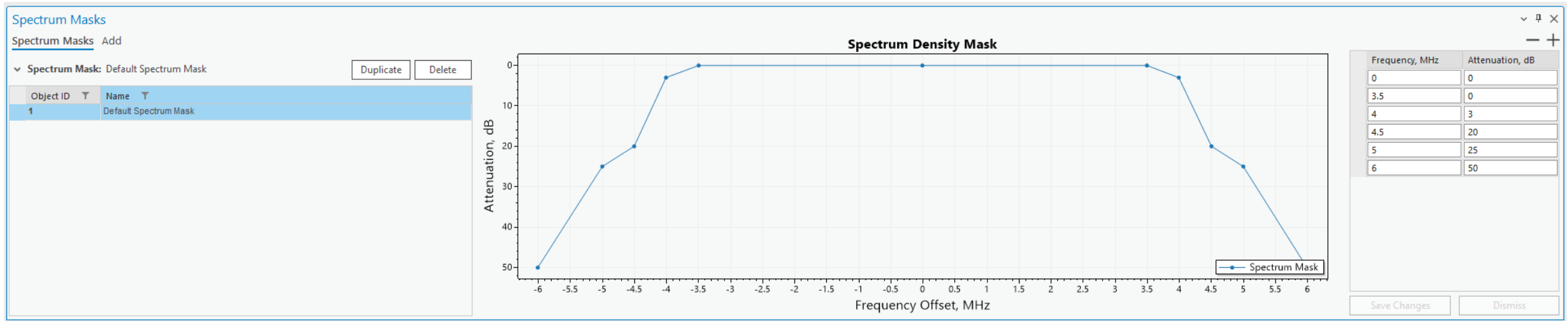
> QAM0512

4

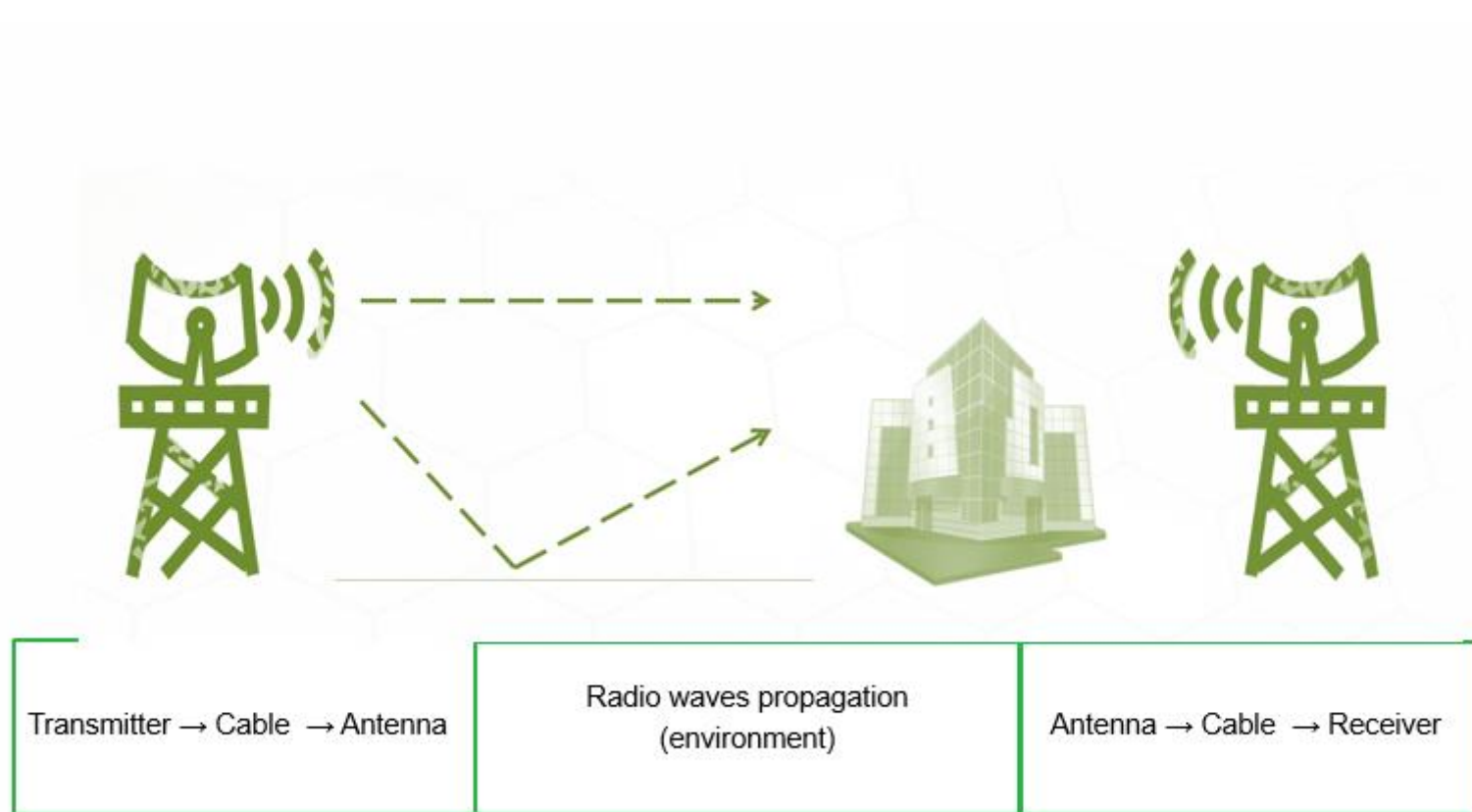
Frequency Plans



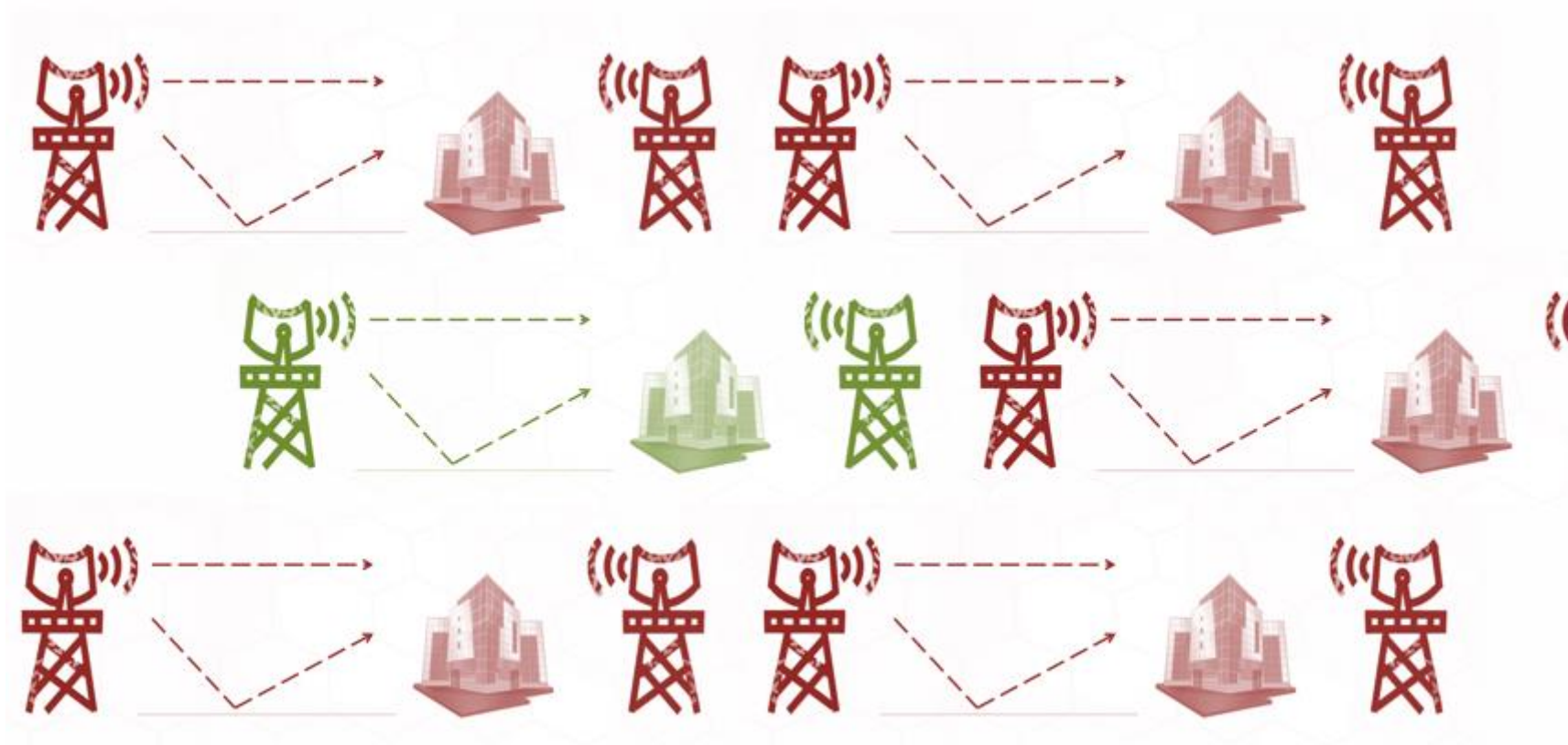
Spectrum Mask



Transmission Network

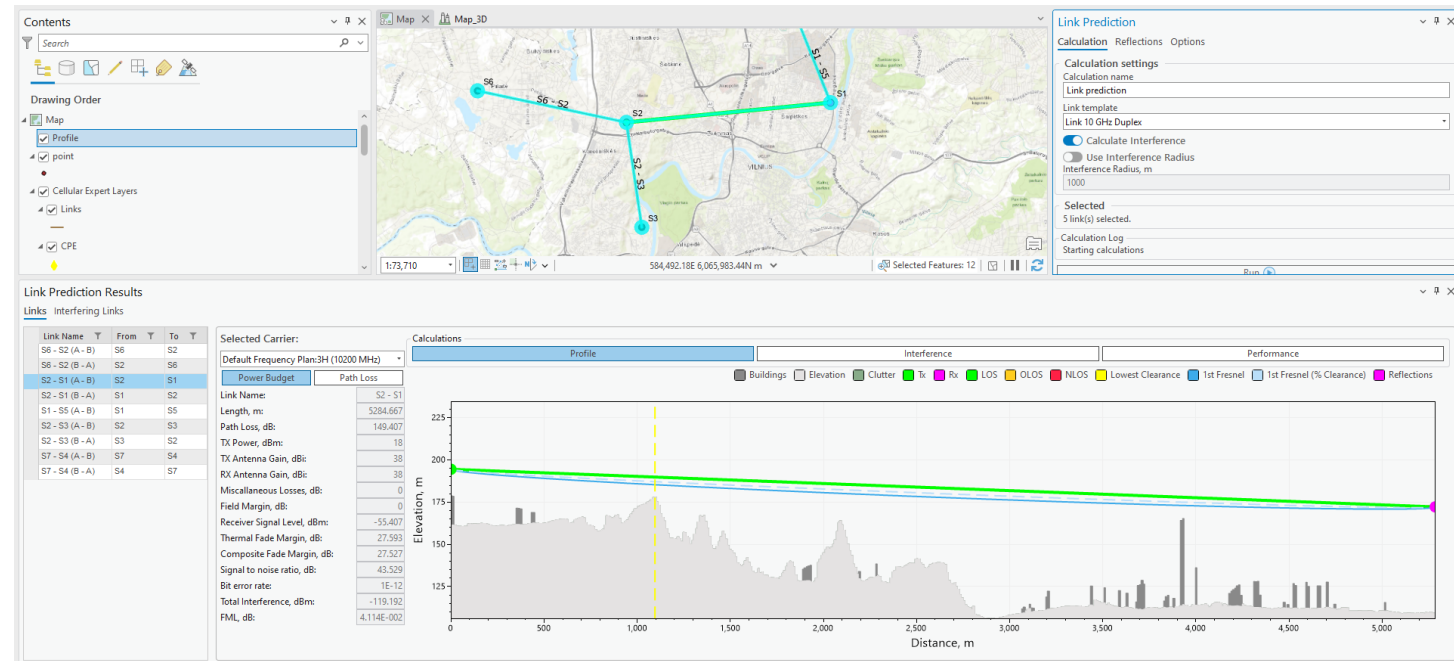


Transmission Network



Microwave link planning

- ☐ Power budget
- ☐ Path loss
- ☐ Profile graphical view
- ☐ Interference From
- ☐ Interference To



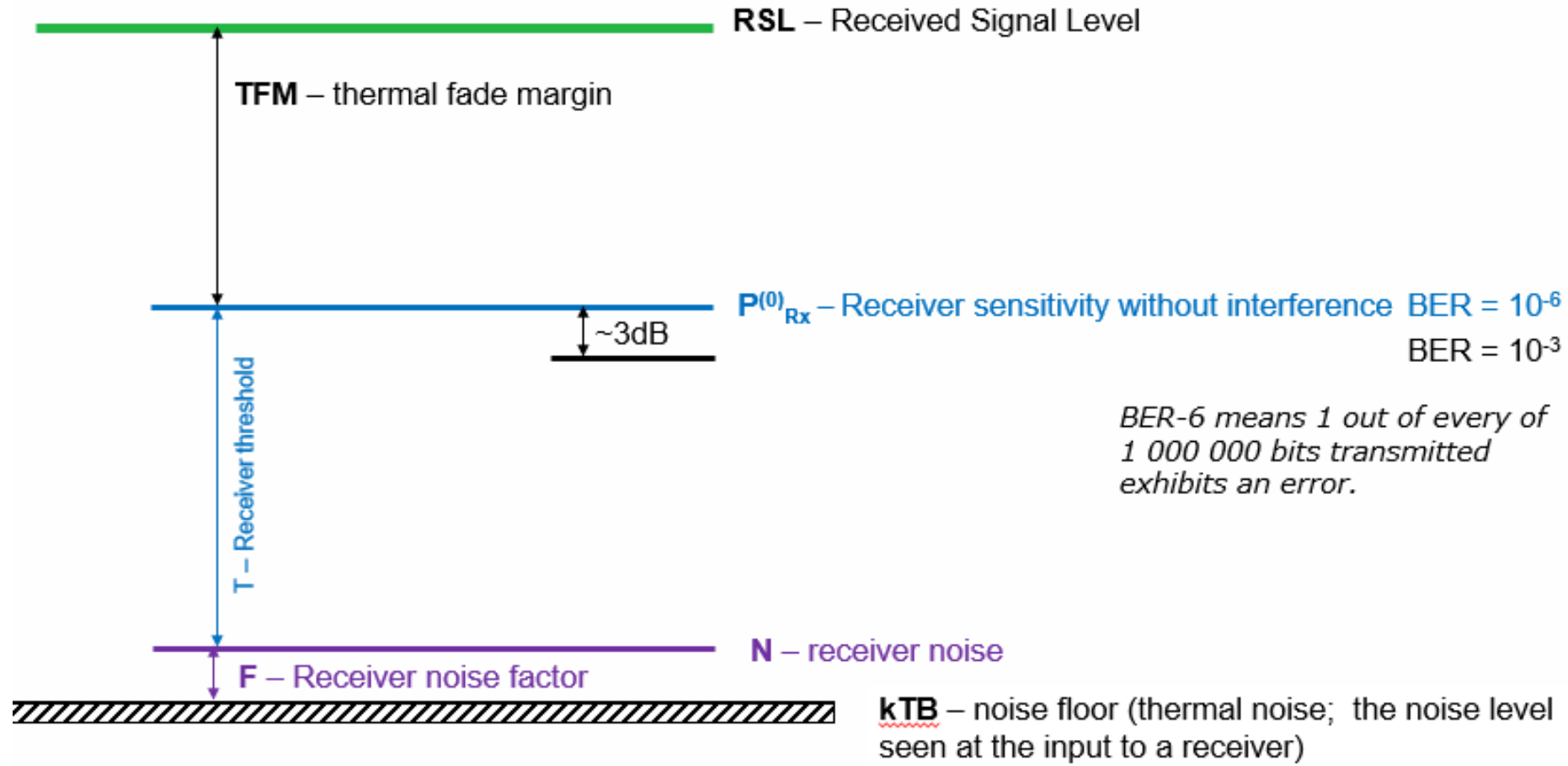
Interference from (interfering links):

Link Name	Site Name	Channel	Length, m	Loss, dB	Interference, dBm	SIR, dB	FML, dB
S6 - S2	S6	Default Frequency Plan:1H	9114.938	159.719	-223.189	167.782	1.647E-012
S6 - S2	S6	Default Frequency Plan:3H	9114.938	159.954	-223.424	168.017	1.560E-012
S6 - S2	S2	Default Frequency Plan:1H	5284.667	147.423	-227.423	172.016	6.210E-013
S6 - S2	S2	Default Frequency Plan:3H	5284.667	147.678	-127.678	72.271	5.854E-003
S2 - S3	S2	Default Frequency Plan:2H	5284.667	147.551	-219.729	164.322	3.653E-012
S2 - S3	S2	Default Frequency Plan:3H	5284.667	147.678	-119.856	64.448	3.533E-002
S2 - S3	S3	Default Frequency Plan:2H	5284.667	147.678	-119.856	64.448	3.533E-002

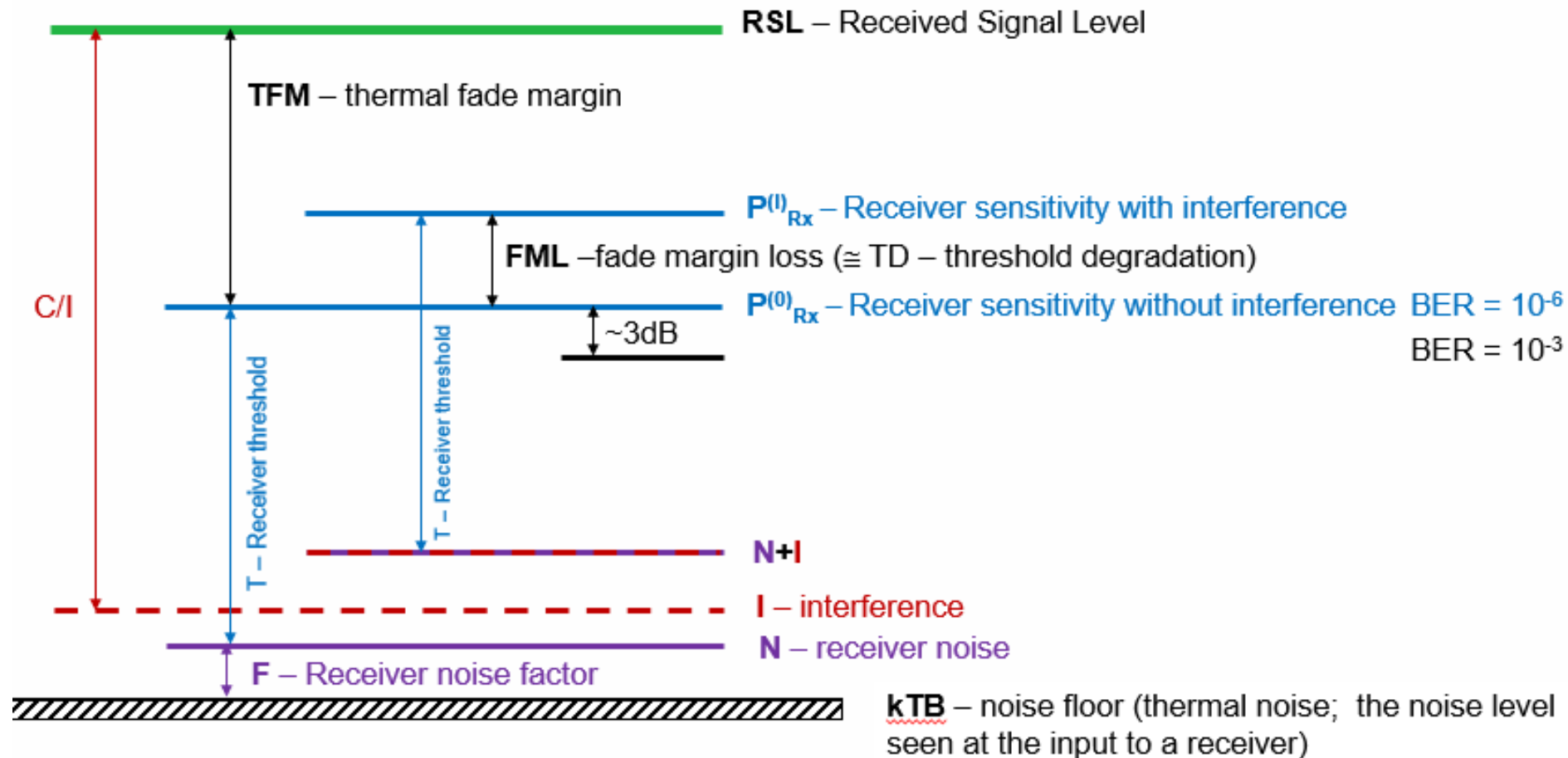
Interference to (interfered links):

Link Name	Site Name	Channel	Length, m	Loss, dB	Interference, dBm	SIR, dB	FML, dB
S2 - S1	S6	Default Frequency Plan:1H	3934.285	124.069	-204.069	174.172	1.345E-010
S2 - S1	S6	Default Frequency Plan:3H	3934.285	124.069	-104.069	74	1.171E+000
S2 - S1	S5	Default Frequency Plan:2H	5867.623	158.382	-254.605	191.606	9.643E-016
S2 - S1	S3	Default Frequency Plan:2H	2737.919	160.446	-232.624	166.372	1.871E-013
S2 - S1	S3	Default Frequency Plan:3H	2737.919	160.446	-132.624	66.244	1.875E-003
S2 - S1	S4	Default Frequency Plan:1H	6809.677	128.835	-227.321	168.57	6.355E-013
S2 - S1	S4	Default Frequency Plan:3H	6809.677	128.835	-227.321	168.57	6.355E-013

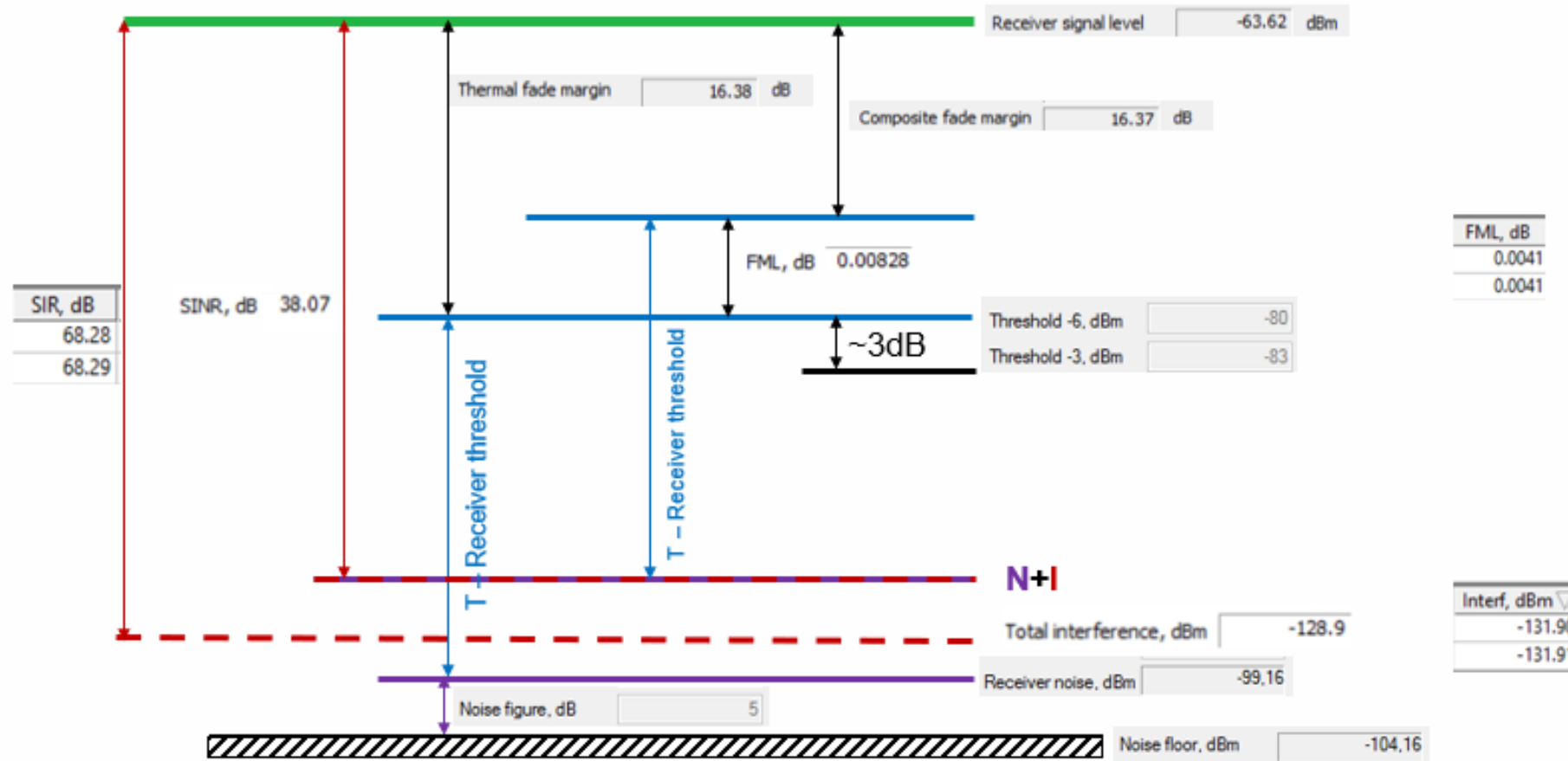
Microwave link planning. Power Budget



Microwave link planning. Power Budget



Microwave link planning. Power Budget



Microwave link planning. Geoclimatic data

Geoclimatic Data

Gaseous Absorption

Temperature

Multipath Fading

Rain Fading

Statistics

Dry air pressure, hPa1013

Water vapour density, g/m37.5

☒ Use geoclimatic data

Water vapour density dataSURFWV

Water vapour density fieldRHO50

Absorption Spectrum

Geoclimatic Data

Gaseous Absorption

Temperature

Multipath Fading

Rain Fading

Statistics

Annual temperature, K:288

☒ Use geoclimatic data

Annual temperature data:ESATEMP

Annual temperature field:TEMP_K

Geoclimatic Data

Gaseous Absorption

Temperature

Multipath Fading

Rain Fading

Statistics

Rain fall data

☐ Rain zone

Category:ITUZone:E

ITU - E

☒ Geoclimatic data

ESA rain rate data:ESARAIN

ESA rain rate field:MT

Fading method

☒ ITU method

ITU-R P.530 version:18

Rain rate exceeded for 0.01% of the time (mm/h)

☐ Use defau22

☒ Use custoi50

☒ Crane

End-to-end reliability method

Multi-hop:Averaging

Two-way:Averaging

Geoclimatic Data

Gaseous Absorption

Temperature

Multipath Fading

Rain Fading

Statistics

☒ ITU method

ITU-R P.530 version:18

☒ Use geoclimatic data

Refractivity gradient data:DNDZ

Refractivity gradient field:DN01

Geoclimatic factor:

☐ Define10

☒ Calculate

Refractivity gradient not exceeded 1% of a year-500N-unit/km

Approximation of terrain roughness:☐ Constant☒ Linear

Percentage of time that refractivity gradient in the lowest 100 m is less than 100 units/km10

C0, dB1.7

CLat, dB0

CLon, dB3

Fading Options:

Use Deep Fading

Percentage of time:

☐ Small percentages of time

☒ All percentages of time

Design:

☐ Detailed

☒ Quick planning

☒ Vigants-Barnett method

Geoclimatic factor (propagation conditions):

☒ 0.5 - good

☐ 2.0 - difficult

☐ 1.0 - average

☐ Other0.5

Worst-month to annual statistics conversion

☒ ITU-R P.530

☐ ITU-R P.841

Interf. correction factor for multipath and focusing effects

Time percentage0.01

Geoclimatic Data

Gaseous Absorption

Temperature

Multipath Fading

Rain Fading

Statistics

ITU-R P.841 statistics conversion

$P_w = Q \cdot P$

P - average annual time percentage of excess

Pw - average annual worst-month time percentage of excess

Q - conversion factor

Conversion factor

☒ Define4

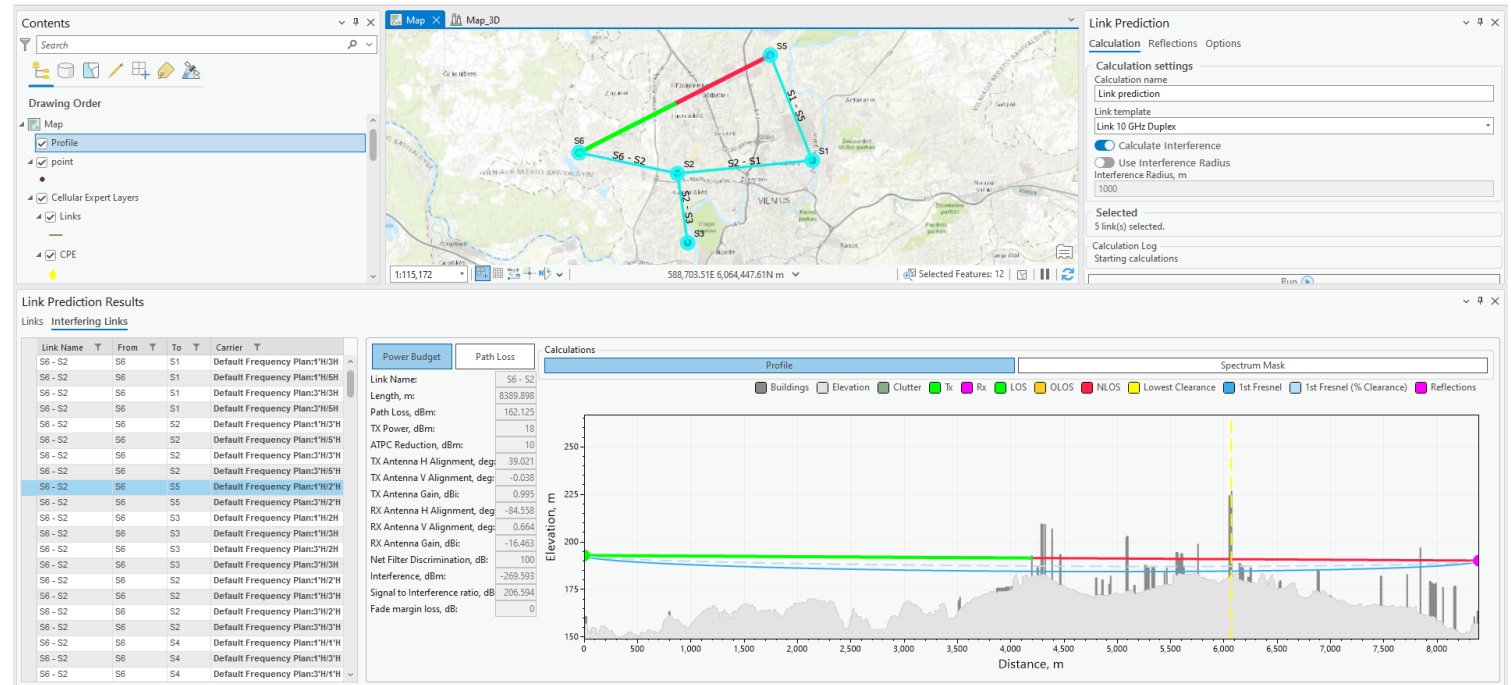
☐ Calculate

Q1 (1...12)2.85

Beta0.13

Microwave link planning: Interfering Links

- ❑ Interfering link on the map
- ❑ Power budget
- ❑ Path loss
- ❑ Profile graphical view
- ❑ Spectrum Mask



Description: C:\CE_Course\0. Descriptions

Name: 7. RL Introduction.pdf



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